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DEPARTMENT OF COMPUTER ENGINEERING

Subject : Machine Learning Lab		Remark																					
Name: _____ Roll No. : _____																							
Class: T.Y. Comp Engg. Batch : T Division: _____		Signature																					
Expt. No. : 07 Date : / /																							
Title : Write a program to implement the Random Forest Algorithm for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets and Visualize model with Python																							
Rubrics :	<table border="1" style="width: 100%; border-collapse: collapse;"><thead><tr><th colspan="3" style="text-align: center;">Date of Submission:</th></tr><tr><th style="width: 33%;">Marks Split up to</th><th style="width: 33%;">Maximum Marks</th><th style="width: 33%;">Marks Obtained</th></tr></thead><tbody><tr><td>Performance/conduction</td><td></td><td></td></tr><tr><td>Report Writing</td><td></td><td></td></tr><tr><td>Attendance</td><td></td><td></td></tr><tr><td>Viva/Oral</td><td></td><td></td></tr><tr><td>Total Marks</td><td></td><td></td></tr></tbody></table>		Date of Submission:			Marks Split up to	Maximum Marks	Marks Obtained	Performance/conduction			Report Writing			Attendance			Viva/Oral			Total Marks		
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Aim :	The main aim of this lab is to demonstrate the how to Creating and Visualizing Random Forest algorithm with Python Secondary aim of this lab is to understand - What is Ensemble Learning? - What is Random Forest and its used for - How to Fit a Random Forest using Scikit-Learn - How to Visualize Random Forest using Matplotlib																						
Software Required :	Jupyter Notebook , Python 3.6, Or Google Colab																						
Dataset Used :	heart.csv																						

Theory :

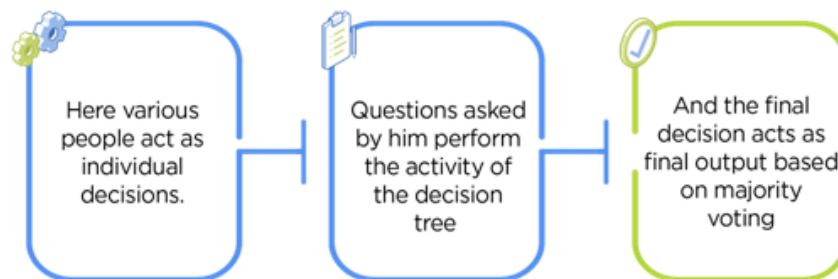
What is Random Forest Algorithm?

Random Forest is one of the most popular and commonly used algorithms by Data Scientists. Random forest is a Supervised Machine Learning Algorithm that is used widely in Classification and Regression problems. It builds decision trees on different samples and takes their majority vote for classification and average in case of regression.

One of the most important features of the Random Forest Algorithm is that it can handle the data set containing continuous variables, as in the case of regression, and categorical variables, as in the case of classification. It performs better for classification and regression tasks. In this tutorial, we will understand the working of random forest and implement random forest on a classification task.

Real-Life Analogy of Random Forest

Let's dive into a real-life analogy to understand this concept further. A student named X wants to choose a course after his 10+2, and he is confused about the choice of course based on his skill set. So he decides to consult various people like his cousins, teachers, parents, degree students, and working people. He asks them varied questions like why he should choose, job opportunities with that course, course fee, etc. Finally, after consulting various people about the course he decides to take the course suggested by most people.

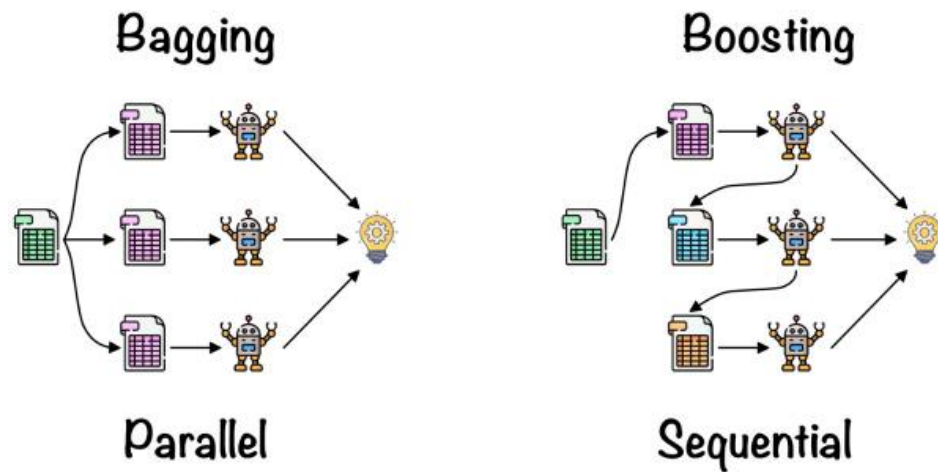


Working of Random Forest Algorithm

Before understanding the working of the random forest algorithm in machine learning, we must look into the ensemble learning technique. Ensemble simply means combining multiple models. Thus a collection of models is used to make predictions rather than an individual model.

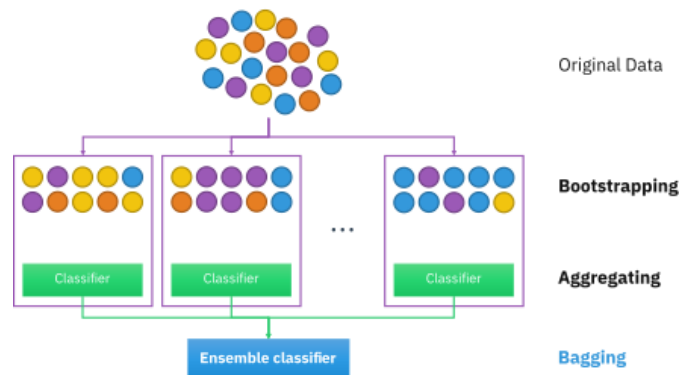
Ensemble uses two types of methods:

- 1. Bagging**– It creates a different training subset from sample training data with replacement & the final output is based on majority voting. For example, Random Forest.
- 2. Boosting**– It combines weak learners into strong learners by creating sequential models such that the final model has the highest accuracy. For example, ADA BOOST, XG BOOST.

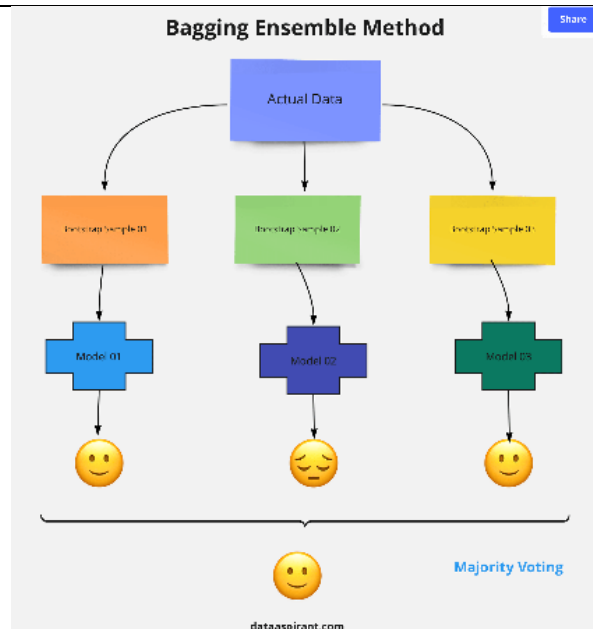


Bagging

Bagging, also known as Bootstrap Aggregation, is the ensemble technique used by random forest. Bagging chooses a random sample/random subset from the entire data set. Hence each model is generated from the samples (Bootstrap Samples) provided by the Original Data with replacement known as row sampling. This step of row sampling with replacement is called bootstrap. Now each model is trained independently, which generates results. The final output is based on majority voting after combining the results of all models. This step which involves combining all the results and generating output based on majority voting, is known as aggregation.



Now let's look at an example by breaking it down with the help of the following figure. Here the bootstrap sample is taken from actual data (Bootstrap sample 01, Bootstrap sample 02, and Bootstrap sample 03) with a replacement which means there is a high possibility that each sample won't contain unique data. The model (Model 01, Model 02, and Model 03) obtained from this bootstrap sample is trained independently. Each model generates results as shown. Now the Happy emoji has a majority when compared to the Sad emoji. Thus based on majority voting final output is obtained as Happy emoji.



Boosting

Boosting is one of the techniques that use the concept of ensemble learning. A boosting algorithm combines multiple simple models (also known as weak learners or base estimators) to generate the final output. It is done by building a model by using weak models in series.

There are several boosting algorithms; AdaBoost was the first really successful boosting algorithm that was developed for the purpose of binary classification. AdaBoost is an abbreviation for Adaptive Boosting and is a prevalent boosting technique that combines multiple “weak classifiers” into a single “strong classifier.” There are Other Boosting techniques. For more, you can visit

Procedure:

Steps Involved in Random Forest Algorithm

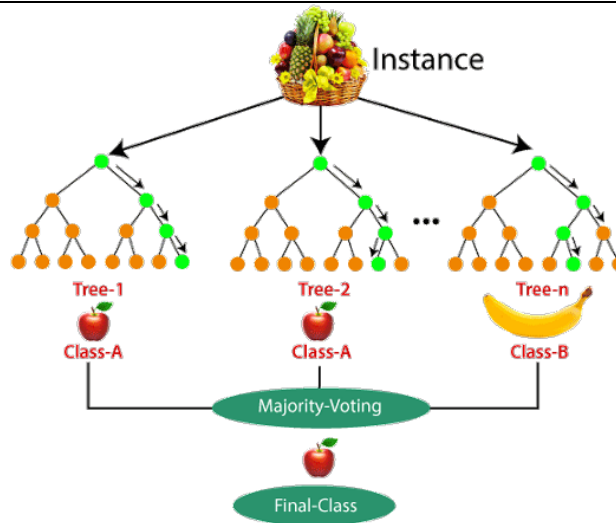
Step 1: In the Random forest model, a subset of data points and a subset of features is selected for constructing each decision tree. Simply put, n random records and m features are taken from the data set having k number of records.

Step 2: Individual decision trees are constructed for each sample.

Step 3: Each decision tree will generate an output.

Step 4: Final output is considered based on Majority Voting or Averaging for Classification and regression, respectively.

For example: consider the fruit basket as the data as shown in the figure below. Now n number of samples are taken from the fruit basket, and an individual decision tree is constructed for each sample. Each decision tree will generate an output, as shown in the figure. The final output is considered based on majority voting. In the below figure, you can see that the majority decision tree gives output as an apple when compared to a banana, so the final output is taken as an apple.



Important Features of Random Forest

Diversity: Not all attributes/variables/features are considered while making an individual tree; each tree is different.

Immune to the curse of dimensionality: Since each tree does not consider all the features, the feature space is reduced.

Parallelization: Each tree is created independently out of different data and attributes. This means we can fully use the CPU to build random forests.

Train-Test split: In a random forest, we don't have to segregate the data for train and test as there will always be 30% of the data which is not seen by the decision tree.

Stability: Stability arises because the result is based on majority voting/ averaging.

Difference Between Decision Tree and Random Forest

Random forest is a collection of decision trees; still, there are a lot of differences in their behavior.

Decision trees	Random Forest
1. Decision trees normally suffer from the problem of overfitting if it's allowed to grow without any control.	1. Random forests are created from subsets of data, and the final output is based on average or majority ranking; hence the problem of overfitting is taken care of.
2. A single decision tree is faster in computation.	2. It is comparatively slower.
3. When a data set with features is taken as input by a decision tree, it will formulate some rules to make predictions.	3. Random forest randomly selects observations, builds a decision tree, and takes the average result. It doesn't use any set of formulas.

Thus random forests are much more successful than decision trees only if the trees are diverse and acceptable.

Important Hyperparameters in Random Forest

Hyperparameters are used in random forests to either enhance the performance and predictive power of models or to make the model faster.

Hyperparameters to Increase the Predictive Power

n_estimators: Number of trees the algorithm builds before averaging the predictions.

	max_features: Maximum number of features random forest considers splitting a node. mini_sample_leaf: Determines the minimum number of leaves required to split an internal node. criterion: How to split the node in each tree? (Entropy/Gini impurity/Log Loss) max_leaf_nodes: Maximum leaf nodes in each tree <i>Hyperparameters to Increase the Speed</i> n_jobs: it tells the engine how many processors it is allowed to use. If the value is 1, it can use only one processor, but if the value is -1, there is no limit. random_state:controls randomness of the sample. The model will always produce the same results if it has a definite value of random state and has been given the same hyperparameters and training data. oob_score: OOB means out of the bag. It is a random forest cross-validation method. In this, one-third of the sample is not used to train the data; instead used to evaluate its performance. These samples are called out-of-bag samples. Advantages and Disadvantages of Random Forest Algorithm Advantages 1. It can be used in classification and regression problems. 2. It solves the problem of overfitting as output is based on majority voting or averaging. 3. It performs well even if the data contains null/missing values. 4. Each decision tree created is independent of the other; thus, it shows the property of parallelization. 5. It is highly stable as the average answers given by a large number of trees are taken. 6. It maintains diversity as all the attributes are not considered while making each decision tree though it is not true in all cases. 7. It is immune to the curse of dimensionality. Since each tree does not consider all the attributes, feature space is reduced. 8. We don't have to segregate data into train and test as there will always be 30% of the data, which is not seen by the decision tree made out of bootstrap. Disadvantages 1. Random forest is highly complex compared to decision trees, where decisions can be made by following the path of the tree. 2. Training time is more than other models due to its complexity. Whenever it has to make a prediction, each decision tree has to generate output for the given input data.
Task to Perform:	1. Implement Random Forest and find the accuracy of your model 2. Visualize your build model in python
Conclusion	In this way we successfully implement and visualize Random Forest in python and find its accuracy

Test Your knowledge

1. Give the Best Estimator using GridSearch CV

2. For the Best Estimator model we created in lab session write the following results

oob_score of Classifier	
Accuracy score of Best Model	
Features of Best Estimator	
Feature Importance	Age = Sex = Cholestrol = BP =

3. For the following Parameter Grid Apply the Grid Search Cv and find Best Estimator

'max_features': list(range(0,14))

'bootstrap': [True, False],

'max_depth': [10, 20, 30, 40, 50, 60, 70, 80, 90, 100, None],

'min_samples_leaf': [1, 2, 4],

'min_samples_split': [2, 5, 10],

'n_estimators': [200, 400, 600, 800, 1000, 1200, 1400, 1600, 1800, 2000]

write a code in your .ipynb file and find following result

oob_score of Classifier	
Accuracy score of Best Model	
Features of Best Estimator	
Feature Importance	Age = Sex = Cholestrol = BP =